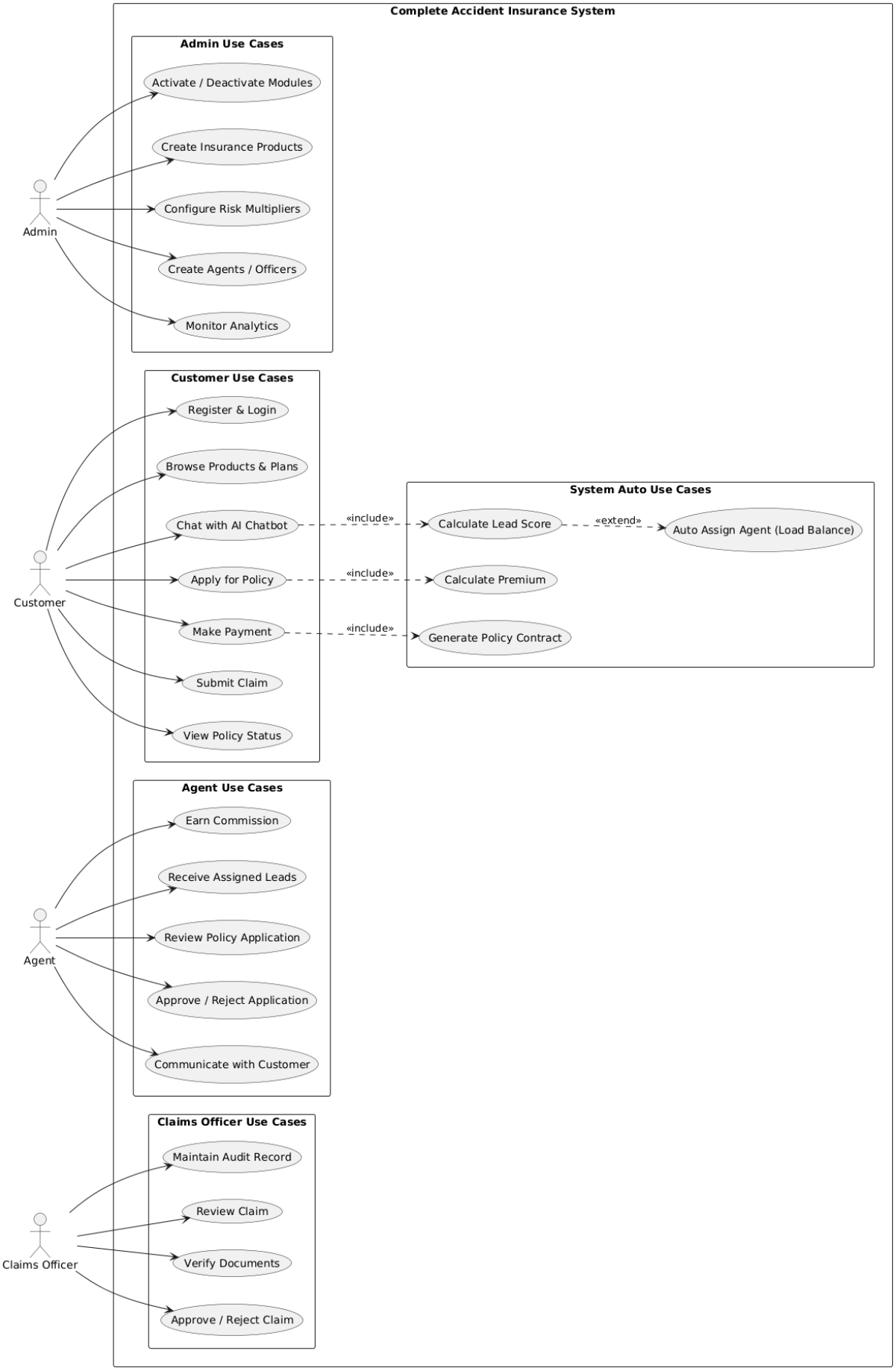
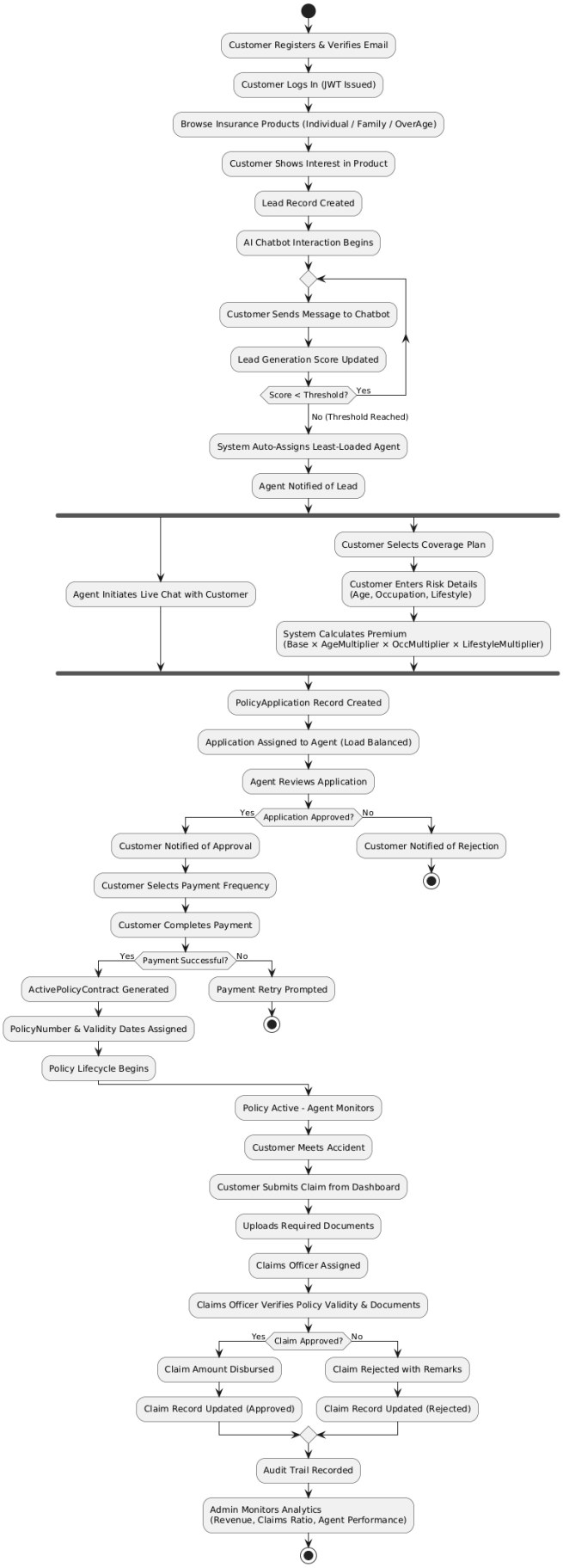
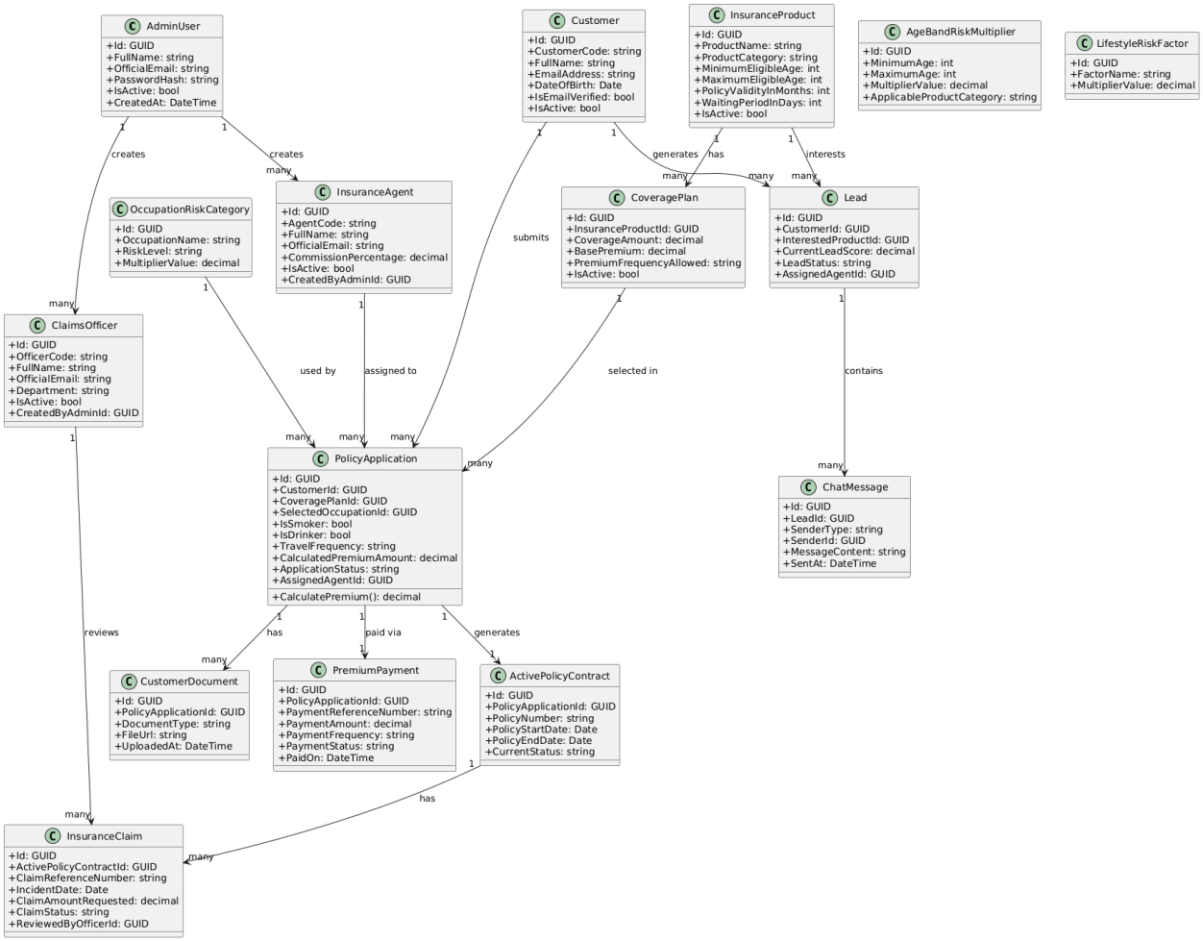

Complete Accident Insurance System
Admin Use Cases
Activate / Deactivate Modules
Create Insurance Products
Configure Risk Multipliers
Create Agents / Officers
Monitor Analytics
Admin
Customer Use Cases
Register & Login
Browse Products & Plans
Chat with AI Chatbot
Apply for Policy
Make Payment
Submit Claim
View Policy Status
Customer
«include»
«include»
«include»
System Auto Use Cases
Calculate Lead Score
«extend»
Auto Assign Agent (Load Balance)
Calculate Premium
Generate Policy Contract
Agent Use Cases
Earn Commission
Receive Assigned Leads
Review Policy Application
Approve / Reject Application
Communicate with Customer
Agent
Claims Officer Use Cases
Maintain Audit Record
Review Claim
Verify Documents
Approve / Reject Claim
Claims Officer

Complete Accident Insurance - End-to-End Activity Flow




AdminUser
+Id: GUID
+FullName: string
+OfficialEmail: string
+PasswordHash: string
+IsActive: bool
+CreatedAt: DateTime
Customer
+Id: GUID
+CustomerCode: string
+FullName: string
+EmailAddress: string
+DateOfBirth: Date
+IsEmailVerified: bool
+IsActive: bool
InsuranceProduct
+Id: GUID
+ProductName: string
+ProductCategory: string
+MinimumEligibleAge: int
+MaximumEligibleAge: int
+PolicyValidityInMonths: int
+WaitingPeriodInDays: int
+IsActive: bool
AgeBandRiskMultiplier
+Id: GUID
+MinimumAge: int
+MaximumAge: int
+MultiplierValue: decimal
+ApplicableProductCategory: string
LifestyleRiskFactor
+Id: GUID
+FactorName: string
+MultiplierValue: decimal
OccupationRiskCategory
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+OccupationName: string
+RiskLevel: string
+MultiplierValue: decimal
InsuranceAgent
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+AgentCode: string
+FullName: string
+OfficialEmail: string
+CommissionPercentage: decimal
+IsActive: bool
+CreatedByAdminId: GUID
CoveragePlan
+Id: GUID
+InsuranceProductId: GUID
+CoverageAmount: decimal
+BasePremium: decimal
+PremiumFrequencyAllowed: string
+IsActive: bool
Lead
+Id: GUID
+CustomerId: GUID
+InterestedProductId: GUID
+CurrentLeadScore: decimal
+LeadStatus: string
+AssignedAgentId: GUID
ClaimsOfficer
+Id: GUID
+OfficerCode: string
+FullName: string
+OfficialEmail: string
+Department: string
+IsActive: bool
+CreatedByAdminId: GUID
PolicyApplication
+Id: GUID
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+CoveragePlanId: GUID
+SelectedOccupationId: GUID
+IsSmoker: bool
+IsDrinker: bool
+TraveFrequency: string
+CalculatedPremiumAmount: decimal
+ApplicationStatus: string
+AssignedAgentId: GUID
+CalculatePremium(): decimal
ChatMessage
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+LeadId: GUID
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+SenderId: GUID
+MessageContent: string
+SentAt: DateTime
CustomerDocument
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+PolicyApplicationId: GUID
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+UploadedAt: DateTime
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+PolicyApplicationId: GUID
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+PolicyEndDate: Date
+CurrentStatus: string
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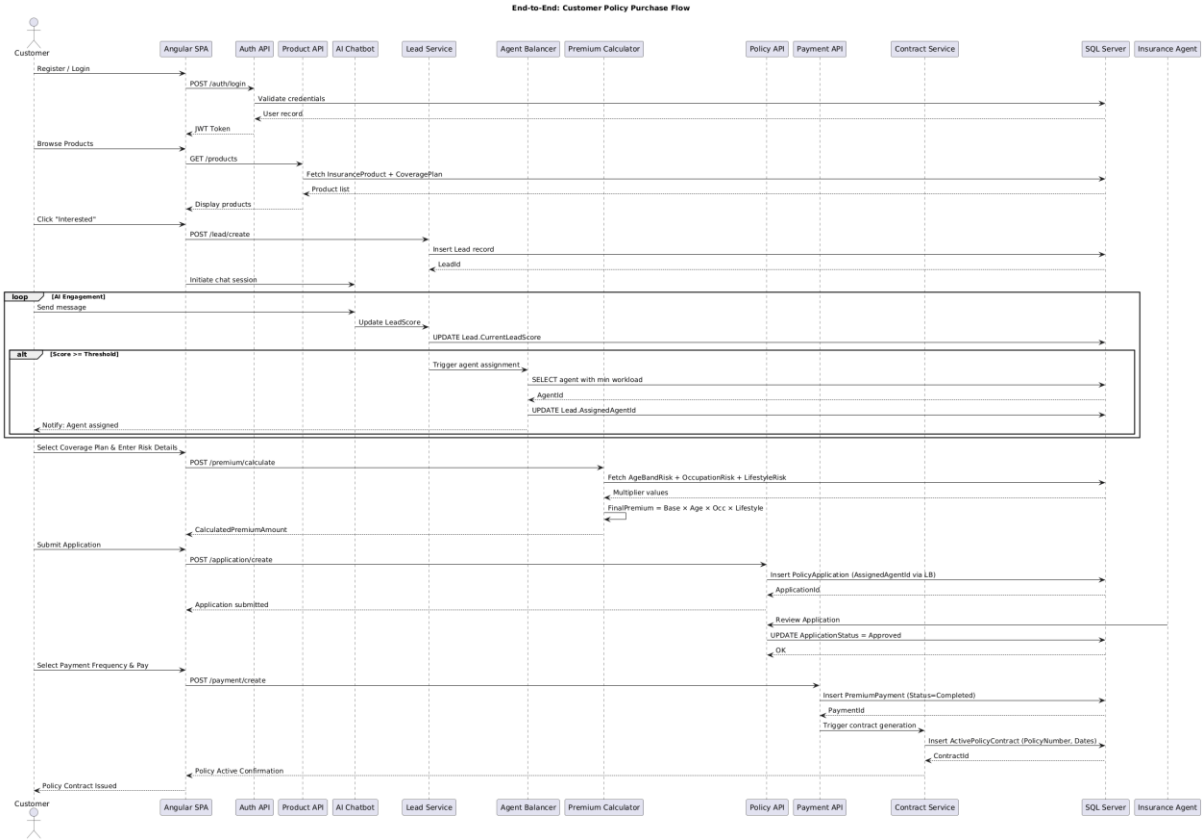
+IncidentDate: Date
+ClaimAmountRequested: decimal
+ClaimStatus: string
+ReviewedByOfficerId: GUID
creates
creates
generates
has
interests
submits
used by
assigned to
selected in
contains
reviews
has
paid via
generates
has


End-to-End: Customer Policy Purchase Flow
Customer
Angular SPA
Auth API
Product API
AI Chatbot
Lead Service
Agent Balancer
Premium Calculator
Policy API
Payment API
Contract Service
SQL Server
Insurance Agent
Register / Login
POST /auth/login
Validate credentials
User record
JWT Token
Browse Products
GET /products
Fetch InsuranceProduct + CoveragePlan
Product list
Display products
Click "Interested"
POST /lead/create
Insert Lead record
LeadId
Initiate chat session
loop [AI Engagement]
Send message
Update LeadScore
UPDATE Lead.CurrentLeadScore
alt [Score >= Threshold]
Trigger agent assignment
SELECT agent with min workload
AgentId
UPDATE Lead.AssignedAgentId
Notify: Agent assigned
Select Coverage Plan & Enter Risk Details
POST /premium/calculate
Fetch AgeBandRisk + OccupationRisk + LifestyleRisk
Multiplier values
FinalPremium = Base × Age × Occ × Lifestyle
CalculatedPremiumAmount
Submit Application
POST /application/create
Insert PolicyApplication (AssignedAgentId via LB)
ApplicationId
Application submitted
Review Application
UPDATE ApplicationStatus = Approved
OK
Select Payment Frequency & Pay
POST /payment/create
Insert PremiumPayment (Status=Completed)
PaymentId
Trigger contract generation
Insert ActivePolicyContract (PolicyNumber, Dates)
ContractId
Policy Active Confirmation
Policy Contract Issued